Machine Vise CAD Modeling and Drawing Package

Parametric part models, detail drawings, and assembly documentation for an eight-component bench vise

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Course	ME 403 Engineering Design, NJIT	**Software**	SolidWorks part, assembly, and drawing environments
Team	Samuel Cohen, Luis Ramirez, Larimel Rosario, Diego Arias, Miguel Huayahuas	**Deliverable**	12-sheet ANSI A drawing set: 9 part sheets, exploded assembly with BOM, 2 assembly section sheets

PROJECT SUMMARY

A bench machine vise was modeled and documented in SolidWorks from a given set of reference drawings. Each of the eight components (base, movable jaw, jaw plates, ACME lead screw, handle rod, handle knobs, machine screws, and keys) was built as a fully constrained parametric part, with selected dimensions driven by the last digits of the student ID. The parts were mated into a working assembly and documented in a 12-sheet drawing package with orthographic, section, detail, and isometric views, a standard NJIT title block on every sheet, an exploded assembly with balloons and a bill of materials, and assembly sections showing how the screw, jaw, and base engage.

1. Introduction

Engineering drawings remain the legal and manufacturing record of a part: a machinist, inspector, or supplier works from the drawing, not the CAD file. The objective of this assignment was to reproduce a complete drawing package for a machine vise so that every feature could be manufactured and inspected from the sheets alone. This required building each part with correct design intent, choosing views that fully define the geometry, and assembling the parts to confirm that they fit and move as intended.

2. Components and Drawing Sheets

Table 1. Bill of materials and drawing sheets in the package

Item	Part	Qty	Key features shown	Scale
1	Vise base	1	Mounting lugs, jaw guideways, threaded boss for the lead screw; full section A-A	2:5
2	Jaw plate	2	Knurled gripping face (0.01 in depth), countersunk screw holes, detail view of the knurl	1:1
3	Machine screw	4	10-24 UNC flat-head screw with slotted head; section through the head	1:1
4	Lead screw	1	ACME 5/8"-8 thread, cross hole for the handle rod	1:1
5	Movable jaw	1	Two sheets: guide slots, screw bore, tapped plate holes, sections A-A and B-B	3:4
6	Handle rod	1	Straight rod passing through the screw cross hole	3:2
7	Handle knob	2	Bored knob with 0.005 in × 45° chamfers; section A-A	3:1
8	Jaw key	2	Rectangular key locating the jaw in the base slots	3:1

3. Approach

Each part was sketched on the plane that best matched its manufacturing datum and dimensioned so that the model would update predictably: symmetric features were mirrored about center planes, hole patterns were located from datum faces, and the thread, knurl, and chamfer features were added last so they did not drive the base geometry. Values marked ## and ### in the reference drawings were set from the last two and three digits of the student ID.

The drawings follow standard third-angle projection practice. Every sheet uses an ANSI A border with the NJIT Department of Mechanical and Industrial Engineering title block, listing the part title, scale, sheet number, file name, and who drew and checked it; group members checked each other’s sheets before submission. Section views were added wherever internal features such as bores, threads, and slots would otherwise be shown with hidden lines, and enlarged detail views were used for small features like the knurl pattern and the ACME thread form.

4. Results

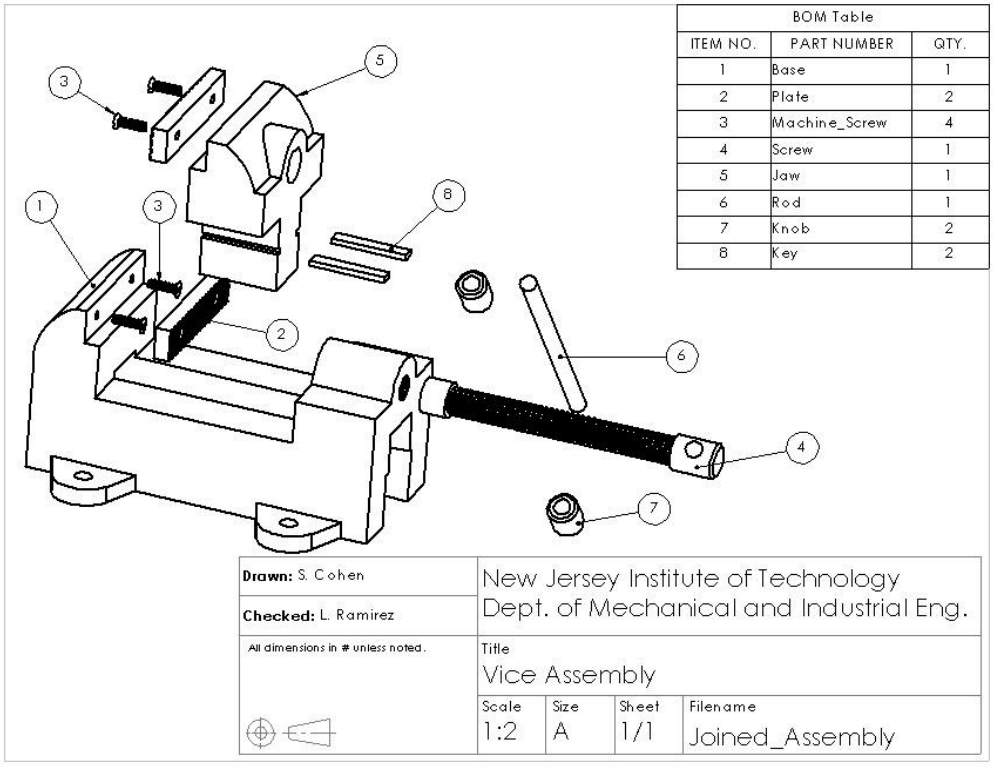


Figure 1. Exploded vise assembly with item balloons and bill of materials.

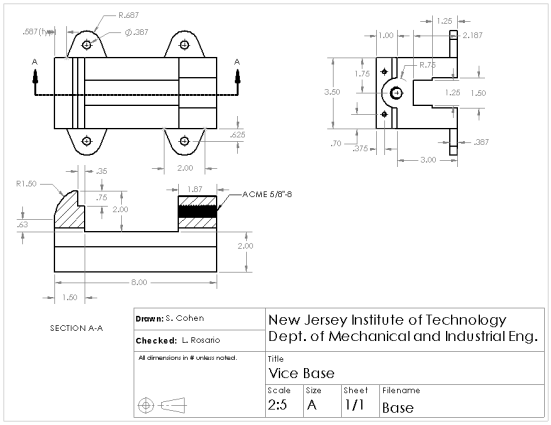


Figure 2. Vise base detail drawing (2:5)

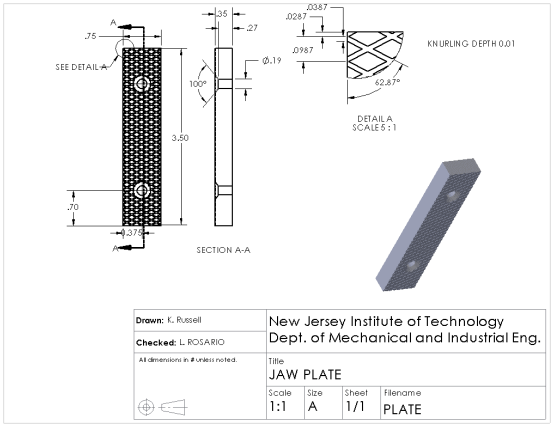


Figure 3. Jaw plate with knurl detail (1:1)

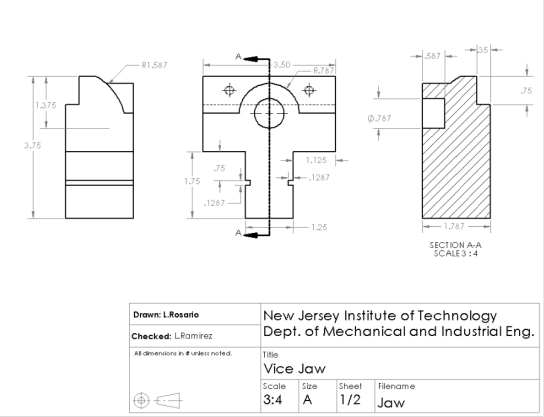


Figure 4. Movable jaw, sheet 1 of 2 (3:4)

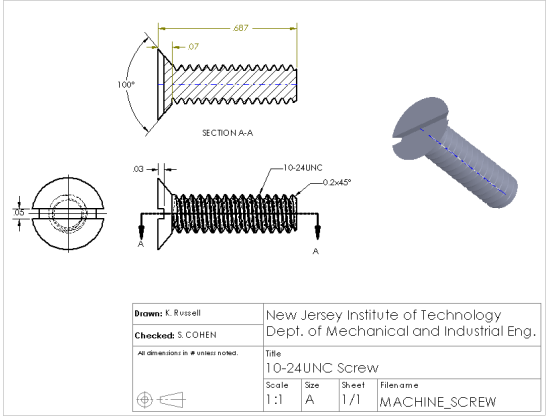


Figure 5. 10-24 UNC machine screw (1:1)

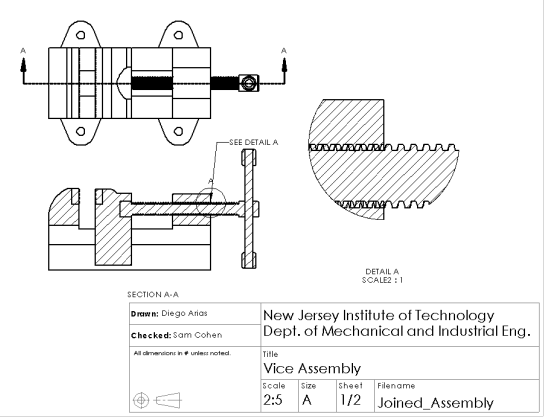


Figure 6. Assembly section through the lead screw and jaw

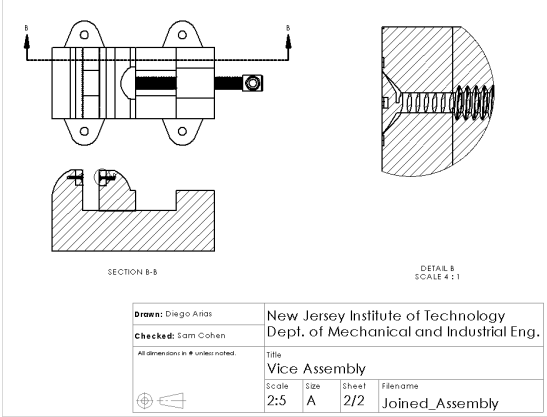


Figure 7. Assembly section with thread detail

5. Discussion

Design intent. Building the parts around their functional datums paid off at assembly: the jaw keys, guide slots, and lead-screw bore lined up without rework, and the vise could be opened and closed through its full travel in the assembly model. Because the ID-driven dimensions were placed as driving values, changing them regenerated the parts and drawings without broken references.

Drawing clarity. The parts with the most internal detail, the base and the movable jaw, needed section views and, for the jaw, a second sheet to keep the views uncluttered. Detail views at enlarged scale were the clearest way to define the knurl depth and the thread profiles, which are too small to dimension legibly at the part scale.

Assembly documentation. The exploded view with balloons tied each component to the bill of materials, and the two assembly section sheets showed the internal engagement of the ACME screw with the base and jaw, which cannot be seen from the outside. Together they let someone unfamiliar with the vise understand how it goes together.

6. Conclusion

A complete, manufacturable drawing package was produced for an eight-component machine vise. The 12 sheets define every part with orthographic, section, and detail views under a consistent title block, and the exploded assembly, bill of materials, and assembly sections document how the vise fits together. The work reinforced dimensioning for design intent, view selection, and drawing checking as a team.

References

ASME. (2018). *Dimensioning and tolerancing* (ASME Y14.5-2018). American Society of Mechanical Engineers.

New Jersey Institute of Technology. (2026). *ME 403 Assignment 1: Drawing reproduction*. Department of Mechanical and Industrial Engineering.